



SAFETY DATA SHEET

QSR POWDER BLEACH

MSDS No.:

AA00280-0920072030

1. PRODUCT AND COMPANY IDENTIFICATION

Product name : 파우더 표백제
QSR POWDER BLEACH

Other means of identification : Not applicable.

Recommended use : Sanitizer

Restrictions on use : Reserved for industrial and professional use.

Product dilution information : 0.09 %

Company : ECOLAB KOREA LTD.
8F, Daemyung Tower
135 Beobwon-ro, Songpa-gu, Seoul Korea 05836
82-1588-7595

Emergency telephone number : 82-1588-7595

Issuing date : 11.03.2021

2. HAZARDS IDENTIFICATION

GHS Classification

Product AS SOLD

Acute toxicity (Inhalation) : Category 4
Serious eye damage/eye irritation : Category 2
Specific target organ toxicity - single exposure : Category 3 (Respiratory system)
Chronic aquatic toxicity : Category 2

Product AT USE DILUTION

Not applicable.

GHS Label element

Product AS SOLD

Hazard pictograms



Signal Word : Warning

Hazard Statements : H319 Causes serious eye irritation.
H332 Harmful if inhaled.
H335 May cause respiratory irritation.
H411 Toxic to aquatic life with long lasting effects.

Precautionary Statements : **Prevention:**
P261 Avoid breathing dust, powder.
P264 Wash the contact area thoroughly after handling.

SAFETY DATA SHEET

QSR POWDER BLEACH

P271 Use only outdoors or in a well-ventilated area.

P273 Avoid release to the environment.

P280e Wear eye protection/face protection.

Response:

P304 + P340 + P312 IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing. Call a POISON CENTER or doctor/ physician if you feel unwell.

P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P337 + P313 If eye irritation persists: Get medical advice/ attention.

P391 Collect spillage.

Storage:

P403 + P233 Store in a well-ventilated place. Keep container tightly closed.

P405 Store locked up.

Disposal:

P501 Dispose of contents and container according to wastes control act.

Product AT USE DILUTION

Hazard pictograms : Not applicable.

Signal Word : Not applicable.

Hazard Statements : Not applicable.

Precautionary Statements : **Prevention:**
P264 Wash hands thoroughly after handling.
Response:
P314 Get medical advice/ attention if you feel unwell.
Storage:
P401 Store in accordance with local regulations.

Product AS SOLD

Other hazards : Mixing this product with acid or ammonia releases chlorine gas.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Product AS SOLD

Pure substance/mixture : Mixture

Chemical Name	Common name	CAS-No.	Concentration: (%)
Sodium Carbonate	no data available	497-19-8	40 - 50
Sodium dichloro-s-triazinetriene dihydrate	no data available	51580-86-0	20 - 24.9

Product AT USE DILUTION

No hazardous ingredients

All unlisted components do not contribute to the overall hazard of the product and are trade secret, making up 100% of the full composition.

4. FIRST AID MEASURES

SAFETY DATA SHEET

QSR POWDER BLEACH

Product AS SOLD

In case of eye contact	: IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
In case of skin contact	: Rinse with plenty of water.
If swallowed	: Rinse mouth. Get medical attention if symptoms occur.
If inhaled	: Remove to fresh air. Treat symptomatically. Get medical attention.
Protection of first-aiders	: If potential for exposure exists refer to Section 8 for specific personal protective equipment.
Notes to physician	: Treat symptomatically.
Most important symptoms and effects, both acute and delayed	: See Section 11 for more detailed information on health effects and symptoms.

Product AT USE DILUTION

In case of eye contact	: Rinse with plenty of water.
In case of skin contact	: Rinse with plenty of water.
If swallowed	: Rinse mouth. Get medical attention if symptoms occur.
If inhaled	: Get medical attention if symptoms occur.

5. FIREFIGHTING MEASURES

Product AS SOLD

Suitable extinguishing media	: Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Unsuitable extinguishing media	: None known.
Specific hazards during firefighting	: Not flammable or combustible.
Hazardous combustion products	: Decomposition products may include the following materials: Carbon oxides nitrogen oxides (NOx) Sulphur oxides
Special protective equipment for firefighters	: Use personal protective equipment.
Specific extinguishing methods	: Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes.

6. ACCIDENTAL RELEASE MEASURES

Product AS SOLD

SAFETY DATA SHEET

QSR POWDER BLEACH

Personal precautions, protective equipment and emergency procedures : Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8.

Environmental precautions : Do not allow contact with soil, surface or ground water.

Methods and materials for containment and cleaning up : Sweep up and shovel into suitable containers for disposal.

Product AT USE DILUTION

Personal precautions, protective equipment and emergency procedures : Refer to protective measures listed in sections 7 and 8.

Environmental precautions : No special environmental precautions required.

Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

7. HANDLING AND STORAGE

Product AS SOLD

Advice on safe handling : Do not ingest. Avoid contact with skin and eyes. Do not breathe dust/fume/gas/mist/vapours/spray. Use only with adequate ventilation. Wash hands thoroughly after handling. Mixing this product with acid or ammonia releases chlorine gas. In case of mechanical malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE).

Conditions for safe storage : Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.

Storage temperature : 0 °C to 50 °C

Product AT USE DILUTION

Advice on safe handling : In case of mechanical malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE).

Conditions for safe storage : Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Product AS SOLD

Components with workplace control parameters

Contains no substances with occupational exposure limit values.

Biological occupational exposure limits : no data available

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

SAFETY DATA SHEET

QSR POWDER BLEACH

Personal protective equipment

Eye protection	: Safety glasses with side-shields
Hand protection	: No special protective equipment required.
Skin protection	: No special protective equipment required.
Respiratory protection	: When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.
Hygiene measures	: Handle in accordance with good industrial hygiene and safety practice. Wash face, hands and any exposed skin thoroughly after handling.

Product AT USE DILUTION

Engineering measures	: Good general ventilation should be sufficient to control worker exposure to airborne contaminants.
----------------------	--

Personal protective equipment

Eye protection	: No special protective equipment required.
Hand protection	: No special protective equipment required.
Skin protection	: No special protective equipment required.
Respiratory protection	: No personal respiratory protective equipment normally required.

9. PHYSICAL AND CHEMICAL PROPERTIES

	Product AS SOLD	Product AT USE DILUTION
Appearance	: powder	liquid
Colour	: opaque, white	colourless
Odour	: Chlorine	Chlorine
pH	: 9.0 - 11.0, (1 %)	9.0 - 11.0
Flash point	: Not applicable., Does not sustain combustion.	
Odour Threshold	: no data available	
Melting point/freezing point	: no data available	
Initial boiling point and boiling range	: > 100 °C	
Evaporation rate	: no data available	
Flammability (solid, gas)	: Not applicable.	
Upper explosion limit	: no data available	
Lower explosion limit	: no data available	
Vapour pressure	: no data available	
Relative vapour density	: no data available	
Relative density	: 1.0 - 1.15	
Water solubility	: soluble	

SAFETY DATA SHEET

QSR POWDER BLEACH

Solubility in other solvents	: no data available
Partition coefficient: n-octanol/water	: no data available
Auto-ignition temperature	: no data available
Thermal decomposition	: no data available
Viscosity, kinematic	: no data available
Explosive properties	: no data available
Oxidizing properties	: Yes
Molecular weight	: no data available
VOC	: no data available

10. STABILITY AND REACTIVITY

Product AS SOLD

Reactivity	: No dangerous reaction known under conditions of normal use.
Chemical stability	: Stable under normal conditions.
Possibility of hazardous reactions	: Mixing this product with acid or ammonia releases chlorine gas.
Conditions to avoid	: None known.
Incompatible materials	: Acids
Hazardous decomposition products	: In case of fire hazardous decomposition products may be produced such as: Carbon oxides nitrogen oxides (NOx) Sulphur oxides

11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

Potential Health Effects

Product AS SOLD

Eyes	: Causes serious eye irritation.
Skin	: Health injuries are not known or expected under normal use.
Ingestion	: Health injuries are not known or expected under normal use.
Inhalation	: May cause respiratory tract irritation. Harmful if inhaled. May cause nose, throat, and lung irritation.
Chronic Exposure	: Health injuries are not known or expected under normal use.

Product AT USE DILUTION

Eyes	: Health injuries are not known or expected under normal use.
------	---

SAFETY DATA SHEET

QSR POWDER BLEACH

Skin	: Health injuries are not known or expected under normal use.
Ingestion	: Health injuries are not known or expected under normal use.
Inhalation	: Health injuries are not known or expected under normal use.
Chronic Exposure	: Health injuries are not known or expected under normal use.

Experience with human exposure

Product AS SOLD

Eye contact	: Redness, Pain, Irritation
Skin contact	: No symptoms known or expected.
Ingestion	: No symptoms known or expected.
Inhalation	: Respiratory irritation, Cough

Product AT USE DILUTION

Eye contact	: No symptoms known or expected.
Skin contact	: No symptoms known or expected.
Ingestion	: No symptoms known or expected.
Inhalation	: No symptoms known or expected.

Toxicity

Product AS SOLD

Product

Acute oral toxicity	: Acute toxicity estimate : > 2,000 mg/kg
Acute inhalation toxicity	: 4 h Acute toxicity estimate : 2.29 mg/l Test atmosphere: dust/mist
Acute dermal toxicity	: no data available
Skin corrosion/irritation	: no data available
Serious eye damage/eye irritation	: no data available
Respiratory or skin sensitization	: no data available
Carcinogenicity	: no data available
Reproductive effects	: no data available
Germ cell mutagenicity	: no data available
Teratogenicity	: no data available
STOT - single exposure	: no data available
STOT - repeated exposure	: no data available
Aspiration toxicity	: no data available

Components

Acute dermal toxicity	: Sodium dichloro-s-triazinetriene dihydrate LD50 rat: > 5,000 mg/kg
-----------------------	---

SAFETY DATA SHEET

QSR POWDER BLEACH

12. ECOLOGICAL INFORMATION

Product AS SOLD

Toxicity

Environmental Effects : Toxic to aquatic life. Toxic to aquatic life with long lasting effects.

Product

Toxicity to fish : no data available

Toxicity to daphnia and other aquatic invertebrates : no data available

Toxicity to algae : no data available

Components

Toxicity to fish : Sodium Carbonate
96 h LC50 *Lepomis macrochirus* (Bluegill sunfish): 300 mg/l

Components

Toxicity to daphnia and other aquatic invertebrates : Sodium Carbonate
48 h EC50 *Ceriodaphnia* (water flea): 213.5 mg/l

Sodium dichloro-s-triazinetriene dihydrate
48 h EC50 *Daphnia*: 0.196 mg/l

Persistence and degradability

Product AS SOLD

Readily biodegradable.

Product AT USE DILUTION

Readily biodegradable.

Bioaccumulative potential

no data available

Mobility in soil

no data available

Other adverse effects

Ozone layer hazard: not applicable

13. DISPOSAL CONSIDERATIONS

Product AS SOLD

Disposal methods : The product should not be allowed to enter drains, water courses or the soil. Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.

Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

SAFETY DATA SHEET

QSR POWDER BLEACH

Product AT USE DILUTION

Disposal methods : Diluted product can be flushed to sanitary sewer.

Disposal considerations : Dispose of in accordance with local, state, and federal regulations.

14. TRANSPORT INFORMATION

Product AS SOLD

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Land transport (KR_DG)

Not dangerous goods

Sea transport (IMDG/IMO)

Not dangerous goods

Special precautions for user : None

15. REGULATORY INFORMATION

National regulatory information

Industrial Safety and Health Act

Harmful Substances Prohibited from Manufacturing

Not applicable.

Korea. Harmful Substances Requiring Permission

Not applicable.

Substances to be controlled (Article 442 of the Regulation on Industrial Health Standards)

Not applicable.

Controlled Substances Subject to Environment Monitoring

Not applicable.

Controlled Substances Subject to Health Examination

Not applicable.

Substances with Workplace Exposure Limits

Not applicable.

Chemicals Control Act

Toxic chemicals

Not applicable.

Authorized chemicals

Not applicable.

Restricted Toxic Chemicals

Not applicable.

Banned Toxic Chemicals

Not applicable.

Toxic Release Inventory Chemicals

Not applicable.

Accident Precaution Chemicals

Not applicable.

SAFETY DATA SHEET

QSR POWDER BLEACH

Dangerous Substances Safety Management Act:

Not Applicable to Dangerous Materials

Waste Management Law

Industrial waste

The components of this product are reported in the following inventories:

United States TSCA Inventory :

All substances listed as active on the TSCA inventory

Canadian Domestic Substances List (DSL) :

All components of this product are on the Canadian DSL.

Australia. Australian Industrial Chemicals Introduction Scheme (AICIS) :

On the inventory, or in compliance with the inventory

New Zealand. Inventory of Chemicals (NZIoC), as published by ERMA New Zealand :

not determined

Japan. ENCS - Existing and New Chemical Substances Inventory :

On the inventory, or in compliance with the inventory

Korea. Korean Existing Chemicals Inventory (KECI) :

On the inventory, or in compliance with the inventory

Philippines Inventory of Chemicals and Chemical Substances (PICCS) :

On the inventory, or in compliance with the inventory

China Inventory of Existing Chemical Substances :

On the inventory, or in compliance with the inventory

Taiwan Chemical Substance Inventory :

On the inventory, or in compliance with the inventory

16. OTHER INFORMATION

Sources of key data used to compile the Safety Data Sheet

Globally Harmonized System of Classification and Labelling of Chemicals (GHS)

IARC: (International Agency for Research on Cancer)

US. National Toxicology Program (NTP) Report on Carcinogens

ECHA List of Publishable Substances Registered

EU HPVCs (High Production Volume Chemicals)

Issuing date : 11.03.2021

Version : 1.1

Prepared by : Regulatory Affairs

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific

SAFETY DATA SHEET

QSR POWDER BLEACH

material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.